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REVIEW ARTICLE

Does Muscle–Tendon Unit Structure Predispose to Hamstring Strain Injury During Running? A Critical Review

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Abstract

Hamstring strain injury (HSI) remains the most common muscle injury in high-intensity running in humans. The majority of acute HSI occur specifically within the proximal region of the long head of biceps femoris and there is a sustained interest among researchers in understanding the factors that predispose to HSI. The present critical review describes the current understanding of biceps femoris long head (BFlh) structural features that might influence strain injury risk. Inter-individual differences in muscle–tendon architecture and interactions, muscle fiber type and region-specific innervation are likely to influence biceps femoris long head injury risk and might inform why some individuals are at an increased risk of sustaining a HSI during running. However, more research is needed, with future studies focusing on prospective data acquisition, improved computer simulations and direct imaging techniques to better understand the relationship between structural features, hamstring muscle function, and injury risk.

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Key Points

Fiber/fascicle lengthening might be greater in the long head of biceps femoris (BFlh) during running, thereby increasing injury risk. However, architectural gearing allows fibers/fascicles to rotate as they shorten/lengthen during contractions, allowing for a greater length change in the muscle relative to the fibers/fascicles, which should provide significant protection against strain injury.

The intramuscular architectural structure of BFlh is non-uniform, with BFlh pennation being greater in the proximal-middle sections compared to the distal and extreme proximal sections. These variations likely result in variations in both stress and strain across the muscle, which increases injury risk.

Depending on the task, different hamstring muscle activation patterns may occur that could lead to shear forces being generated between compartments within BFlh, which could increase injury risk.

1 Introduction

Maximal-effort running in humans requires high levels of positive muscular power to be produced by agonist muscles in the lower limb. However, these muscles also produce large negative powers to brake limb segments at specific phases of the running cycle. During the production of negative power, some muscles show an increased injury incidence. Of all lower limb injuries, muscle strain injury to the hamstring muscles is particularly prevalent [1]. Among several others, common risk factors for hamstring strain injury (HSI) include increasing age, previous injury, strength imbalances, poor hamstring and quadriceps flexibility or compliance, reduced hip extension, leg length differences and poor intermuscular recruitment patterns [2]. Furthermore, Timmins et al. [3] found that professional soccer players with short biceps femoris fascicle lengths (< 10.6 cm) had fourfold higher injury risk than players with longer fascicles. Isolated injury to the long head of biceps femoris (BFlh) is reported as the most common acute HSI in intermittent running sports and most often occurs during explosive actions such as sprint running (51%) [1, 4]. This finding suggests that anatomical, morphological or physiological factors specific to BFlh may influence injury risk. Injuries to BFlh show a greater involvement of the proximal region compared to the distal region [5–7], with the proximal musculotendinous junction (MTJ) being reported as the most common injury location in young adults [5, 6, 8, 9], suggesting a possible anatomical influence. It is also suggested that fatigue could play a major role in the HSI mechanism(s) [10]. However, the majority of current modelling studies do not test for the influence of fatigue and it is challenging to account for the anatomical variability between individuals and the behavior and influence of the surrounding tissues [10]. The main purpose of this critical review is to consider the potential effects of BFlh anatomical and morphological structure (including origin/insertion points, muscle architecture, tendinous tissue inscriptions, fiber type, and innervation patterns) and the muscle's likely functional role in sprint running on muscle strain injury risk by comparing it to neighboring (synergist) muscles. Understanding these basic muscle–tendon unit risk factors may allow us to identify individuals at risk and allow the development of new hypotheses as to the factors that might reduce hamstring injury risk. We also provide recommendations for future research, summarized in the Supplementary File.

1.1 Anatomical Background

The hamstring muscles form the posterior aspect of the thigh and comprise the semimembranosus (SM), semitendinosus (ST) and the two-headed biceps femoris (BF) [6]; long (BFlh) and short head (BFsh). The biarticular SM, ST and BFlh cross the hip and knee joints [6, 11] and are therefore considered to function both as knee flexors and hip extensors [11]. BFsh is monoarticular and only crosses the knee joint [6] and, therefore, is considered to primarily function as a knee flexor [11]. When the knee is partially flexed, the medial and lateral hamstring muscles have secondary functions, assisting in internal (SM and ST) and external (BFlh and BFsh) rotation of the leg [6, 11]. When investigating the hamstring muscles, some discrepancies have been observed between studies. SM, ST and BFlh originate at the ischial tuberosity [6, 11] and a common proximal tendon for BFlh and ST arising from the medial facet of the ischial tuberosity is reported [12]. However, connections with the sacrotuberous ligament have also been described [13, 14], while some ST muscle fibers have been observed to originate directly from the ischial tuberosity [14]. Relatively lateral or anterolateral to the conjoint BFlh/ST tendon, SM also originates at the ischial tuberosity [9, 15]. A connection of the origin of SM with the BFlh/ST tendon has been reported [9], but not confirmed in other studies [13, 14]. BFsh, the only monoarticular muscle, generally originates from the lateral lip of the inferior third of the linea aspera and supracondylar ridge of the femur [11]. Three insertion sites for BFlh have been reported, including the head of the fibula, the lateral condyle and the lateral collateral ligament [6], while BFsh shares the distal tendon of BFlh for its attachment [6, 11].

ST inserts into the medial aspect of the proximal tibia via the pes anserinus, while SM divides distally into several parts: a direct attachment to the posterior aspect of the medial condyle, a part that blends with the popliteal fascia, and a reflected part that reinforces the intercondylar part of the joint capsule of the knee as the oblique popliteal ligament [6, 11]. At this time, it is not clear whether BFlh origin influences its injury risk, even though its line of action, and thus the route of internal force transmission, might vary depending on this origin.

The observed between-study differences in origin-insertion points reflect significant inter-individual anatomical variations that are commonly reported throughout the body. The reason for the higher injury incidence in BFlh is not currently known, and this lack of understanding contributes to difficulties in identifying individuals at greatest risk of injury as well as the formulation of strategies to both prevent and treat the injury. The present critical review will evaluate our current understanding of BFlh structural features and consider their potential influence on injury predisposition.

2 Musculotendinous Factors That May Predispose Biceps Femoris Long Head (BFlh) to Strain injury

2.1 Muscle–Tendon Architecture and Interactions

Muscle–tendon architecture is defined as the organization of the muscle and tendinous tissues relative to the line of action of the muscle–tendon unit [16]. It has been suggested that variations in muscle–tendon architecture may lead to a decrease in the normal glide and flexibility of the hamstring muscles, which may predispose certain individuals to injury [6]. With regards to the force-transmitting tendinous tissues, Woodley and Mercer [14] reported both long proximal and distal tendons for BFlh and SM (including their aponeuroses) with their proximal and distal tendons overlapping. Thus, the middle sections of BFlh and SM have attachments to both the proximal and distal tendon while this is not the case for ST [9]. It has been hypothesized that this structure might be a predisposing factor to HSI since the proximal SM tendon proceeds distally with a twist before ending as a wide tendinous sheet [9, 14]. Van der Made et al. [9] speculated that this twist might cause the muscular force direction to misalign with the line of action of the tendon, increasing its vulnerability to injury. However, further study is needed to investigate the dynamic interaction of the muscle–tendon–bone complex since it is likely that not only individual muscle characteristics but also the dynamic interaction between proximal tendons, predisposes to muscle injury (e.g. tendons twisted around each other may increase the lever arm during contraction) [9]. Also, a tendinous inscription in ST (raphe) has been reported, which may contribute to the low frequency of injury observed in this muscle relative to BFlh as it could play a role in protecting ST against gross injury [9]. However, future studies are required to elucidate the role of the raphe in potential injury patterns.

Each hamstring muscle also exhibits different architectural features [14, 17–22]. SM and BFlh generally have a larger physiological cross-sectional area and pennation angle [18, 23], while ST and BFsh appear to have smaller pennation angles (note that pennation angle is related to physiological cross-sectional area, which increases with pennation if muscle volume remains constant) [9, 18, 19, 23]. Based on structure–function relationships for muscle [24–26] it therefore seems that SM and BFlh are designed for greater force production and stiffness (shorter fascicles and larger pennation angle) and ST and BFsh for larger excursion and greater shortening velocity (longer fascicles and smaller pennation angle) [18]. Speculatively, this might increase strain injury risk because the fiber/fascicle lengthening per muscle length change might be greater in BFlh [16]. However, the proximal and middle BFlh regions have large angles and long fascicles compared to distal regions [19] yet are more commonly injured during running [5–7]; i.e. it is not possible to make a generalized case for whole-BFlh architecture because it varies substantially along with the muscle (i.e. fascicle curvature and muscle anatomical cross-sectional area differ between sections). Additionally, fibers/fascicles also rotate as they shorten/lengthen during muscle contractions (known as 'architectural gearing'; see Fig. 1), allowing a greater length change in the muscle relative to the fibers/fascicles. This has been clearly shown during muscle lengthening contractions [27] and should provide significant protection against strain injury. The effect is greater in muscles that are highly pennate [27] (such as BFlh), which means individuals with greater BFlh pennation might be expected to have greater protection. Indeed, it has been previously shown that gearing can differ between muscles within a synergistic group, and between concentric and eccentric contractions [28]. Of interest is that longer BFlh fascicles, but not greater pennation, have so far been found to correlate with BFlh injury risk [3, 29], and increases in BFlh fascicle length without increases in pennation have been observed after periods of eccentric strength training, a change thought to confer injury protection [30]. This makes theoretical sense if the longer fascicles contain more sarcomeres in series (and hence fascicle lengthening occurs partly through increases in serial sarcomere number), since less sarcomere strain would then be expected to occur for a given fascicle, and muscle, length change [31]. However, these results were obtained from only one part of the muscle, and fascicle length estimates required significant (> 60%) extrapolation because the image width captured by the

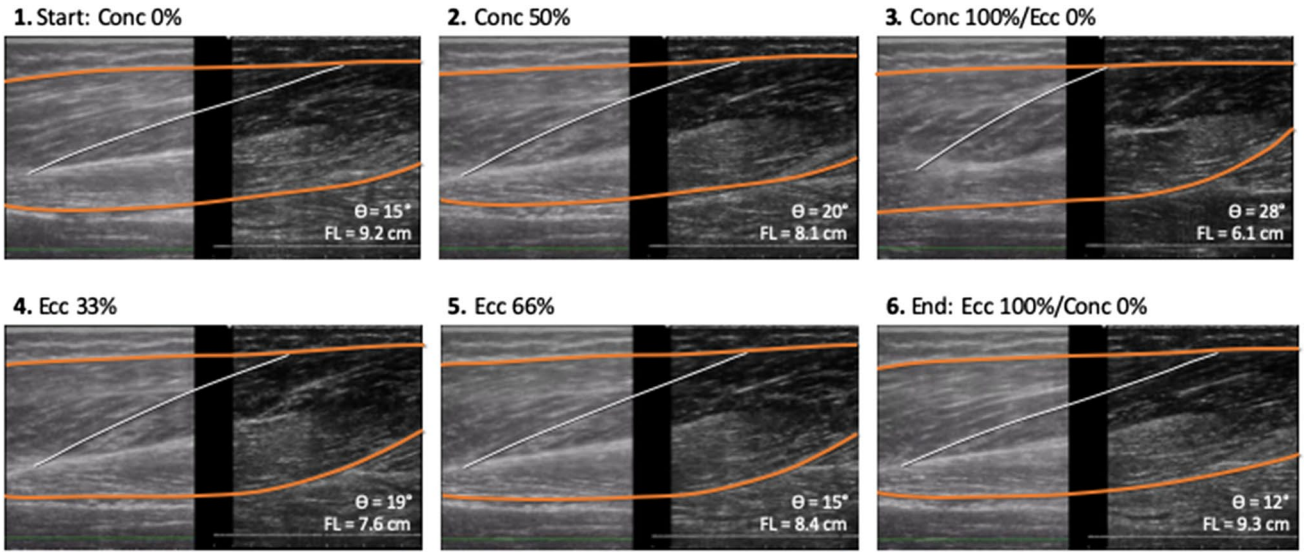

Fig. 1 Long head of biceps femoris (BFlh) muscle belly thickness, fascicle length and fascicle angle each change during concentric and eccentric contractions (through 90° knee flexion at 60% maximum voluntary contraction, in this image series). The significant fascicle rotation increases the architectural gear ratio of the muscle, allowing for a lesser fascicle length change relative to muscle length change, and a theoretical reduction in fiber strain injury risk. Percentages refer to movement time in each phase. The 1-cm between-probe (inter-transducer) space is shown as a vertical black line between images (no image is available in this region). Image acquisition with permission and thanks to Clarissa Brusco, Universidade Federal do Rio Grande do Sul (Brazil) and Edith Cowan University (Australia). *FL* fascicle length, *conc* concentric, *ecc* eccentric

sonography technique was small compared to the length of fascicles; this may induce errors in fascicle length measurements [32]. Furthermore, it is unclear how these estimates of both fascicle length and angle, made in relaxed muscle, might be influenced by small changes in pressures exerted by neighboring muscles, which vary between individuals and change significantly with training [33]. Even light pressures exerted on relaxed muscles can significantly affect fascicle length and angle estimates. Alternatively, inter-individual differences in resting fascicle length, or change (increases) after periods of exercise training, may be explained by factors other than sarcomere addition, including changes in resting sarcomere length, decreases in resting muscle tension, increases in end-point tension (e.g. through increased stiffness of tendon/aponeurosis), or the longitudinal translocation of myofibrils or whole fibers within the muscles [34]; and these may reflect adaptations that could influence injury risk (e.g. extracellular matrix remodeling). Therefore, it is presently unclear whether the resting muscle architecture is indicative of architecture, and gearing, during dynamic muscle contraction. Further examination of these relationships using higher-accuracy measurement techniques will be required in the future in order to more clearly determine the relationship between muscle architecture and injury risk. In the future, diffusion tensor imaging techniques might address some of the limitations of current (ultrasound) imaging techniques; however the existing literature demonstrates that this technique is not ready to be implemented in clinical practice [35]. In terms of injury prevention, it is of interest to further explore the discrepancy between the observed decrease in pennation angle after injury-minimizing eccentric hamstrings training and the hypothesis that greater pennation, allowing a greater fascicle rotation during contraction, might otherwise minimize injury risk.

One issue with the exploration of relationships between the intramuscular architectural structure of muscles such as BFlh and ST and injury risk is that muscle architectures are visibly non-uniform [19]. BFlh has longer fascicles and larger pennation at its proximal and middle regions compared to the distal site [17] (Fig. 2) whilst ST has longer fascicles and larger pennation distally than proximally [19]. Kellis et al. [19] have linked these proximo-distal differences in BFlh architecture with higher excursions in the proximal as compared to the distal region, while the opposite would be true for ST. However, accurate modelling of fascicle behaviors to support this theory is not yet possible in complex muscles such as the hamstrings. As Fig. 2 shows, the fascicle path of BFlh is concave in the proximal region, shifts to a complex curvature in the middle, and becomes convex towards the distal region. Based on anatomical arrangement, it seems that BFlh pennation angle is greater in the proximal-middle sections compared to the distal, and extreme proximal, sections. If and when technology allows it, direct fascicle strain and rotation measurements during sporting and other activities might be possible to determine whether strains in BFlh are, or are not, greater than in its

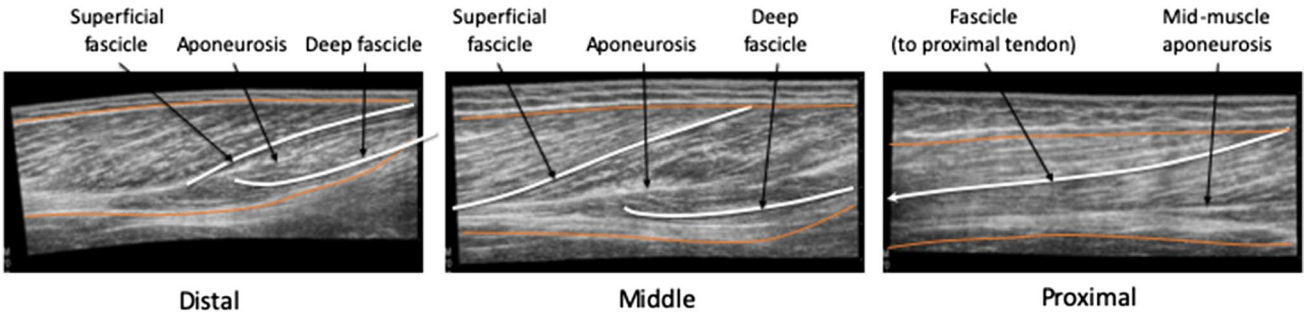

Fig. 2 Extended field-of-view sonograms of the long head of biceps femoris (BFlh), showing that the fascicle path is concave in the proximal region (~33% distance from greater trochanter), exhibits complex curvature in the middle, and becomes convex towards the distal region (~67% of the distance from greater trochanter)

synergist muscles, and whether the magnitude of strain is then associated with an increased injury risk. For example, real-time ultrasound methods that allow more accurate and high-resolution imaging under these circumstances are a current focus of numerous research groups [36]; however, these methods are yet to be broadly used.

Both tendon and aponeurosis properties might also impact on muscle injury rates. With regard to tendons, animal experiments have clearly established the role of tendons as 'shock absorbers', i.e. energy buffers, during rapid eccentric muscle–tendon unit loading [37, 38]. This role is performed by tendons absorbing energy (performing negative work) and thus reducing both the magnitude and rate of muscle fascicle strain. In humans, this mechanism has been observed to reduce or even prevent muscle/fascicle stretch during high-power eccentric tasks such as drop jumps and jump landings [39–41]. For tendons of similar construct (i.e. Young's modulus), tendon length is a critical factor influencing compliance; i.e. compliance is proportional to length. The tendon:fiber length ratio of SM is ~1.5 × that of BFlh and 3 × that of ST [16], so the contribution of tendon length change to muscle–tendon unit length change would be greater for SM followed by BFlh and ST [22, 42]. It follows that SM may have a greater protection than BFlh and ST through the tendon buffer mechanism; however it is not yet clear whether the tendon:fiber ratio of BFlh contributes to its injury prevalence; prospective studies assessing the effect of tendon:fiber ratio, or changes in it, may shed light on this issue.

With regard to the role of the aponeurosis, Kellis et al. [19] hypothesized that the variability in muscle architecture would have implications for the way muscle fibers contract in relation to the aponeurosis and the whole muscle–tendon unit. Specifically, for the same relative shortening of fascicles the aponeurosis displacement will be different in different regions of the muscles [24]. This might speculatively influence injury risk because either (a) fibers in some muscle regions may elongate substantially and thus be at greater risk of injury during muscle stretch, or (b) inconsistencies in strain magnitudes between compartments within the muscle may result in shear forces between fascicles or fibers at the intersection of those compartments, predisposing to tissue disruption. BFlh fibers originate on a relatively narrow proximal aponeurosis and insert on a broader distal aponeurosis [14]. A previous modelling study demonstrated that a smaller ratio of aponeurosis width to muscle width resulted in a larger maximum peak local strain, with the largest observed peak local strain in muscle tissue adjacent to the proximal aponeurosis [43]. Therefore, a larger muscle and/or narrower proximal aponeurosis would theoretically make a muscle more susceptible to injury, which implies that individuals with this architectural pattern may be at greater risk [43].

Computational models predict increases in peak local fiber strain, fiber strain non-uniformity and the amount of muscle tissue that undergoes strain in BFlh at faster sprinting speeds, which likely contributes to injury susceptibility [44]. These models also predict that the muscle tissue near the proximal MTJ experiences a higher stress (due to the smaller cross-sectional area), and thus undergoes a greater strain than neighboring tissue [44]. Moreover, higher levels of activation at faster running speeds may intensify this effect because of the muscle's nonlinear force–length relationship, such that the tissue near the proximal MTJ may experience a disproportional increase in strain [44]. Li and Wang [45] have suggested that this fiber strain and strain rate non-uniformity at fast running speeds could lead to longitudinal muscle injury (caused by shear along the inter-fiber border), tissue separation, and misalignment of myofibrils along the muscle fibers. The dimensions of BFlh relative to the other hamstring muscles could contribute to its greater injury susceptibility during high-intensity running actions, and inter-individual BFlh muscle–tendon architectural differences could influence inter-individual differences in injury risk, although more investigation (including prospective studies) is needed.

2.2 Muscle Fiber Type

Early research examining muscle fiber composition led researchers to speculate that the relatively high proportion of type II fibers in the hamstring muscles compared to other thigh and leg muscles (quadriceps and adductor magnus) might be a predisposing factor for injury [46] based on the capability of type II fibers to produce more tension at a greater rate compared to type I fibers [6] and the greater susceptibility of type II fibers to eccentric-induced muscle damage (EIMD) [47]. Structural differences between muscle fiber types [48] are a possible factor with type II fibers characterized by narrower and weaker Z disks, and stronger sarcomeres thought to be able to pull the weaker ones apart during muscle lengthening contractions, resulting in Z-line streaming [49]. The higher sensitivity of type II fibers to EIMD [47] may partly explain why training interventions designed to increase eccentric hamstring strength [50] decrease HSI risk. It is suggested that the lower fatigue resistance of type II than type I fibers may explain preferential type II fiber muscle damage. During fatiguing exercise, specifically type II fibers are unable to generate sufficient ATP, resulting in a state of rigor that leads to mechanical disruption [48, 51]. In addition, the hamstrings show greater fatigue relative to their antagonists the quadriceps, when assessed during simulated soccer (measured by evaluating peak torque of the hamstrings and quadriceps on an isokinetic dynamometer before and after a simulated soccer exercise) [52]. Taking this into account, together with retrospective [53, 54] and prospective [55] evidence, there is an increased interest in enhancing the strength-endurance capacity of the hamstring muscles [56, 57] as a potential risk reduction strategy.

Given the aforementioned arguments, it is of interest to investigate BFlh fiber type composition in relation to injury risk. Recent research evaluated BFlh composition in vivo (muscle samples taken mid-section ~ 50% thigh-length) and its relationship with knee flexor maximal and explosive strength (isometric, concentric and eccentric peak torque) in a cohort of young healthy individuals. The investigators observed a balanced muscle fiber composition (type I: 47.1 ± 9.1%: type II: 52.9 ± 9.1%) [58]. Moreover, they did not find a greater proportion of type II fibers in the BFlh relative to other muscles (quadriceps and adductor magnus) and muscle fiber composition was not significantly related to knee flexor function [58]. Based on the overall results, it was concluded that the composition of BFlh appears not to explain the high incidence of strain injuries within BFlh [58]; however, the authors noted that the high inter-individual variability in fiber type composition, with some having > 65% type II fibers, may theoretically increase injury risk in some individuals [58]. Future prospective studies are needed in order to determine whether a relationship exists between fiber type (or fiber type of BFlh relative to other hamstring muscles) and injury incidence. In such studies, it might be prudent to obtain samples from multiple sites along the muscle(s) given that Garrett et al. [46] reported a considerable variability in fiber types between muscle regions in a study of cadavers (type II fibers in BFlh: proximal site (PS) = 55.2 ± 7.0% vs. distal site (DS) = 53.8 ± 4.9%; type II fibers in ST: PS = 54.6 ± 4.5% vs. DS = 60.4 ± 12.1%; type II fibers in SM: PS = 51.1 ± 0.4.0% vs. DS = 50.5 ± 6.8%). Given the specific characteristics of different fiber types, these variations between and within the hamstring muscles could result in differing force development profiles and fatigue rates in different regions within the muscles, and might explain the inconsistent findings in relation to hamstring fiber type composition. In theory, increasing overall strength-endurance capacity partly through a shift in fiber type composition may reduce injury risk. One possible consequence of strength-endurance training could be that maximal sprint performance could conceivably be reduced, although maximal running speed is affected by a very large number of variables and many muscles contribute to whole-body power output, so the effect of a small change in one muscle (or muscle group) may be small. This effect could be monitored through regular athlete testing. At present, a clear conclusion cannot be made as to the possible influence of fiber type on injury risk, and research of this nature is unlikely to progress in elite sport since direct fiber type measurements are highly invasive and are unlikely to be done (especially at multiple sites within multiple muscles). Non-invasive fiber type estimation methods might therefore be necessary in order to allow for further investigation [59].

2.3 Region-Specific Muscle Innervation

Region-specific muscle innervation is another potential factor relevant to injury susceptibility. The differential innervation of BF, with BFlh being innervated by the tibial division of the sciatic nerve and BFsh being innervated by the common peroneal division of the sciatic nerve, may result in asynchrony in the coordination or intensity of stimulation of the two heads and has been suggested to contribute to an increased injury risk [6]. Furthermore, Woodley and Mercer [14] found BFlh to be innervated by one single motor nerve ($n = 6$ specimens), which originated from the sciatic nerve in four specimens and from the tibial nerve in two. In three specimens, the nerve divided into two primary branches relatively proximally, with a branch supplying each region of the muscle. Multiple terminal branches were observed in all nerve branches supplying BFlh [14]. In another cadaveric study ($n = 50$), An et al. [60] assessed the location of the motor entry points (where the nerve branch first pierces the muscle belly) of the hamstring muscles and identified two different branching patterns for BFlh [60]. The

branching patterns were classified into type 1 (one primary motor branch issued from the sciatic nerve) and type 2 (two primary motor branches), with incidences of 82% and 18%, respectively [60]. Although the investigators observed two different innervation types, all BFlh motor entry points were found in close proximity and their location was calculated as one group [60]. More specifically, dividing BFlh into *x*- and *y*-coordinates, these motor entry points were found to be located at 41.5 ± 9.4% (141 ± 33 mm) in the *x*-coordinate (proximo-distal direction) and 36.3 ± 10.8% (42 ± 11 mm) in the *y*-coordinate (medio-lateral direction) [60]. Seidel et al. [61] also reported two innervation patterns for BFlh; however, in contrast to An et al. [60], they found that the type 2 pattern was dominant (66%) [61]. Early research by Sunderland et al. [62] noted that 10 of 20 specimens belonged to type 1, eight specimens to type 2 and, interestingly, another two belonged to another type with three primary branches [62]. Branch and Anz [63] also reported the existence of an extended insertion area at the tibia [63]. This would have a direct influence on hamstring function, hamstring muscle architecture and motor patterning [64]. It seems that some fibers within the muscle would act to extend the knee joint and others to flex it due to their respective insertion areas [64]. Cumulatively, these findings suggest the presence of inter-individual differences in the innervation patterns of the hamstring muscles as well as variation in the inputs to different regions of the muscles, although their relationship with injury risk has yet to be studied. Furthermore, recent research has revealed differences in inter-muscular and proximal–distal electromyographic patterns of hamstring muscles during various hamstring exercises, where different exercises showed distinct regional activity patterns both in ST and BFlh [65]. It might therefore be speculated that different regions of the muscle could be activated independently, which may lead to shear forces occurring between compartments within BFlh, potentially causing injury [66]. However, this relationship to injury risk has not yet been explicitly examined. Future studies implementing EMG array electrodes across multiple muscles may be helpful in ascertaining relationships between activation patterns, changes in patterns with training and detraining, and effects of training exercises with specific activation patterns, on HSI risk.

3 Conclusion

Isolated injury to BFlh remains the most common acute hamstring injury and inter-individual differences in BFlh structural features, as summarized in Fig. 3, could clarify why BFlh is most often injured during running and why some individuals are at an increased risk of injury. Based on structure–function relationships, it seems that SM and BFlh are generally designed for relatively greater force production and stiffness (shorter fascicles and larger pennation angle) and ST and BFsh for larger excursion and greater shortening velocity (longer fascicles and smaller pennation angle). These structural differences might increase strain injury risk because the fiber/fascicle lengthening per muscle length change might be greater in BFlh. However, architectural gearing allows fibers/fascicles to rotate as they shorten/lengthen during muscle contractions, which allows for a greater length change in the muscle relative to the fibers/fascicles and should provide significant protection against strain injury. Furthermore, the intramuscular architectural structure of BFlh is not uniform and it seems that BFlh pennation is greater in the proximal-middle sections compared to the distal, and extreme proximal, sections. Direct fascicle strain and rotation measurements during sporting and other activities will be needed in the future to determine whether the magnitude of strain in BFlh is greater than in other muscles and associated with an increased injury risk. In addition, a larger muscle and/or narrower proximal aponeurosis would theoretically make a muscle more susceptible to injury, which implies that individuals with this architectural pattern may be at greater risk. Variations in fiber types in the hamstring muscles could predispose to different fatigue rates and force development changes in different regions within the muscles, which could theoretically contribute to a higher injury risk. However, at present, a clear conclusion cannot be made as to the possible influence of fiber type on injury risk. Inter-individual differences in the innervation patterns of the hamstring muscles may provide some evidence that different regions of the muscle could be activated independently. Depending on the task, different hamstring muscle activation patterns may occur that could lead to shear forces occurring between compartments within BFlh, increasing injury risk. Understanding these risk factors may allow for clearer identification of individuals at greatest risk, possibly allowing for the provision of additional injury preventative training programs. Insight into these factors is also an imperative for the subsequent development of targeted exercise strategies to provide the required protection. At present, it is neither clear which anatomical or morphological factors most strongly influence injury risk, nor how different training strategies (e.g. strength training using specific ranges of motion, movement speeds or contraction modes, motor control optimization, flexibility training, etc.) might generate specific adaptations. For example, it is known that factors highlighted in the present review, including muscle pennation (fascicle angle), fascicle length, aponeurosis width, fiber type and regional muscle activation, may be influenced by training; however, in most cases it is unclear how to best target these factors, and it is not known whether

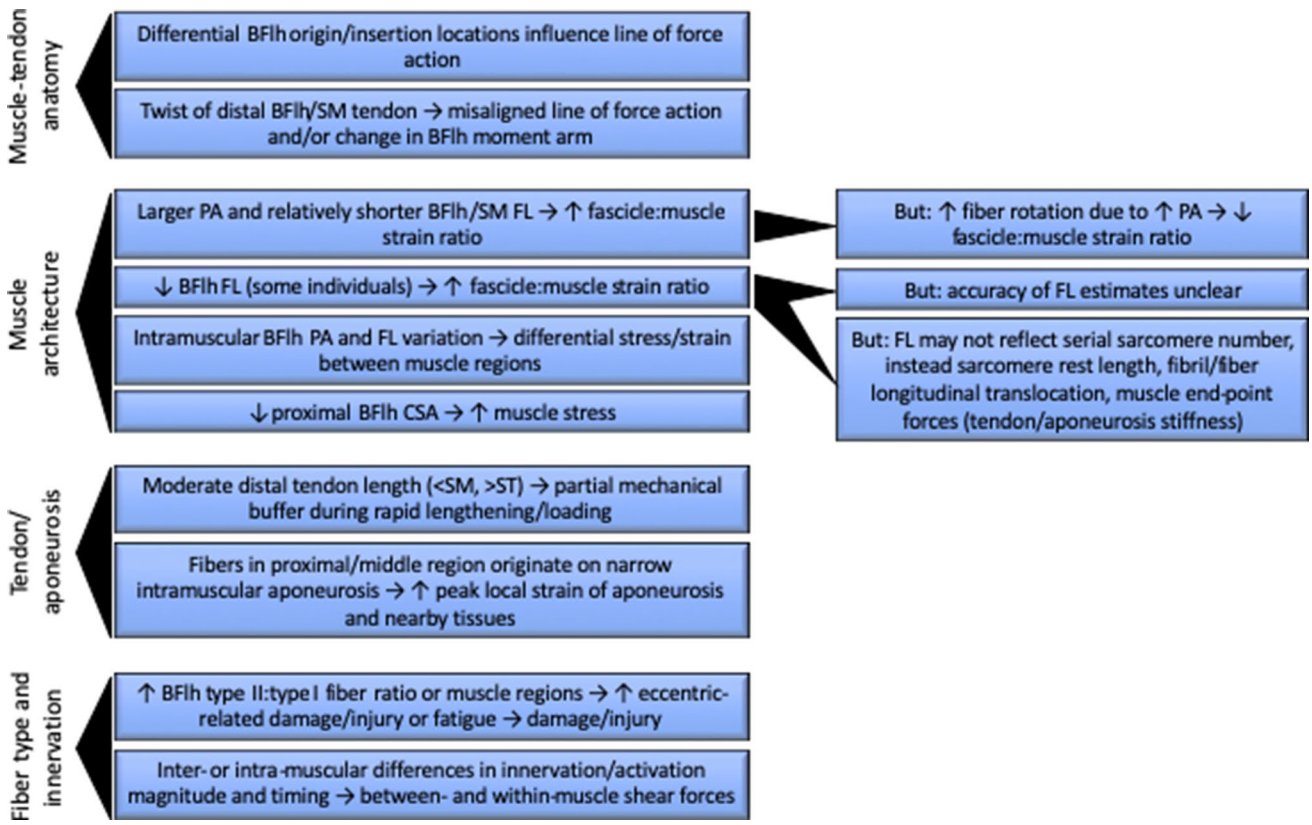

Fig. 3 Anatomical features that may differ within BFlh, between hamstrings components, or between individuals, that may speculatively predispose to BFlh injury risk during running. Whilst a significant research effort has examined many of these hypotheses, there is currently insufficient evidence to confirm any single hypothesis. The provision of additional evidence, subsequent to further advances in musculoskeletal imaging and modelling techniques, is an aim of ongoing research. *BFlh* biceps femoris long head, *SM* semimembranosus, *ST* semitendinosus, *CSA* cross-sectional area (muscle), *FL* fascicle length, *PA* pennation angle

changes directly influence dynamic muscle behavior and subsequent injury risk. Most clearly, strength training interventions have broadly shown a potential to reduce injury risk (e.g., eccentric [67], isometric [68] or strength-endurance [56] biased training); however, it remains to be determined which underlying mechanisms are responsible for this injury risk reduction and whether specific training parameters (contraction mode, load, movement speed, range of motion) might differentially influence this risk between individuals. Future research that focuses on the completion of prospective studies, in addition to improving simulation models and direct imaging techniques, is warranted in order to better understand the relationship between structural features, hamstring muscle function, training interventions, and injury risk. Our recommendations for future research are summarized in the Supplementary File.

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Declarations

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Author contributions SH drafted the article. FC, DC, JC-G, RP, PA and AB revised and contributed to the manuscript according to their respective fields of expertise. All authors read and approved the final manuscript.

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